

ARNA GHOSH

Computer Science, McGill University & MILA

Montréal, Canada

Email ◊ GitHub ◊ LinkedIn ◊ Scholar ◊ Twitter

EDUCATION

McGill University, Montréal

Ph.D. Computer Science

September 2019 - Present

McGill University, Montréal

M.Sc. Neuroscience

September 2017 - August 2019

CGPA: 4.0/4.0

Indian Institute of Technology, Kharagpur

B.Tech. Electrical Engineering

Minor in Computer Science

Micro-specialization in Embedded Wireless Technology

July 2013 - May 2017

CGPA: 9.26/10.0

INTERNSHIPS AND WORK EXPERIENCE

CBrain, McGill Centre for Integrative Neuroscience

July 2019 - August 2019

- Designed a well-documented **Python** wrapper library around the CBrain web platform to run neuroimaging tasks on a high-performance cluster from a UNIX terminal

MITACS Globalink Research intern, McGill University

May 2016 - July 2016

Mechanism underlying sensorimotor integration during motor learning

Dr. Marie-Hélène Boudrias

- Designed the software and hardware interface for tasks to study motor learning characteristics under different external stimulation conditions using **Python**, **C++** and **OpenCV**
- Interfaced EEG, EMG and fMRI setups to the designed software for different experiments. [Media]

Summer intern, University of British Columbia Okanagan

May 2015 - July 2015

Slide-Scanning Microscopy on a Smart Phone by High-Frame-Rate Video Capture

Dr. Kenneth Chau

- Developed the image processing module for fast deblurring of microscopic images extracted from video and developed an integrated hardware-software interface using an Arduino

PROJECTS

Deep Mouse

Doctoral project

Dr. Blake Richards

Sep 2019 - Present

- Working on biologically-plausible deep learning models of the mouse visual cortex.
- Exploring the representation similarity of **Contrastive predictive coding** networks to mice visual cortex and comparing them to standard object recognition deep networks.

Predictive Coding: a variational perspective

Course project

Sep 2019 - Present

- Exploring computational frameworks of predictive coding using **variational autoencoders** in **PyTorch**. Focusing on using egomotion as a cue for predicting the next visual input.

Deep Learning for Neuroimaging data analysis

Sep 2017 - Aug 2019

Masters thesis project

Dr. Marie-Hélène Boudrias & Dr. Georgios D. Mitsis

- Developed a DL framework in **Torch** to identify exercise-induced task-EEG signatures.
- Developed **ccCAM** (cue-combination for Class Activation Map) to improve network interpretability.
- Designed a ML-based **brain age prediction** system from MRI and MEG recordings in **PyTorch**.
- Developed a **Graph-CNN** approach for analyzing functional connectivity data to predict brain age and compared to other ML-based approaches.

Brexting: Brain Texting

Dec 2017 - Aug 2018

Intel Innovate FPGA Grand Finalists

- Built a DL framework on an Intel FPGA board for decoding of imagined movements from EEG recordings to enable real-time typing.
- Won the **Silver award** in regional finals and **Iron award** in grand finals. Code available on repo

Deep Learning for mitotic figure detection

Jul 2016 - May 2017

Bachelor's Thesis Project Dr. Debdoot Sheet

- Built a DL framework to localize and identify mitotic nuclei from *breast histopathological* images on **Torch** and extended the idea to *leukemia detection*.
- Awarded the **Systems Society award** for best project work on Systems.

IndicView

Oct 2014 - Dec 2015

Google-IIT Pilot Project

Dr. Pawan Goyal

- Implemented a binarization algorithm to segregate text from background and extract words from a document written in an Indian language. Accepted to be integrated into Google Translate.

Autonomous Ground Vehicle Research Group

Mar 2014 - May 2017

Perception Team

Dr. Debashish Chakravarty

- Developed image processing modules to detect obstacles, detect lanes and construct a world map by fusing data from camera and LIDAR on **Robot Operating System (ROS)**
- Worked on efficient road segmentation for off-road conditions and visual odometry problems using Convolutional Neural Networks

PUBLICATIONS

Full Papers and Preprints:

1. A. Xifra-Porxas*, **A. Ghosh***, G.D. Mitsis, M.H. Boudrias, *Estimating brain age from structural MRI and MEG data: Insights from dimensionality reduction techniques*, BiorXiv, Nov 2019. [Link](#)
2. **A. Ghosh**, F. Dal Maso, M. Roig, G.D. Mitsis, M.H. Boudrias, *Unfolding the effects of acute cardiovascular exercise on neural correlates of motor learning using Convolutional Neural Networks*, Frontiers in Neuroscience 13 (2019): 1215. [Link](#)
3. E. Renda, S.A. Karmali, I. Yordanova, S. Schwartz, Y. Mahdid, **A. Ghosh**, S. Blain-Moraes, M.H. Boudrias, *Effect of Transcranial Direct Current Stimulation on an Individuals Ability to Learn to Control a Brain-Computer Interface*, McGill Journal of Medicine, October 2019. [Link](#)

*Equal contribution

4. **A Ghosh**, S Singh, D Sheet, *Simultaneous Localization and Classification of Acute Lymphoblastic Leukemic Cells in Peripheral Blood Smears Using a Deep Convolutional Network with Average Pooling Layer*, IEEE 12th International Conference on Industrial and Information Systems (ICIIS), Dec 2017. [Link](#)
5. V Mohanty, S Agrawal, S Datta, **A Ghosh**, V D Sharma, D Chakravarty, *DeepVO: A Deep Learning approach for Monocular Visual Odometry*, ArXiv, Nov 2016 [Link](#)

Poster Presentations:

1. A.Kemtur*, **A. Ghosh***, Arthur Dehgan, Tarek Lajnef, Raphael Vallat, Jean-Baptiste Eichenlaub, Perrine Ruby, Karim Jerbi, *Unraveling the neural signatures of dream recall in EEG: a deep learning approach*, Montreal Artificial Intelligence & Neuroscience, November 2019.
2. **A. Ghosh***, A. Xifra-Porxas*, G.D. Mitsis, M.H. Boudrias. *Combining structural MRI images and MEG recordings for Biological brain age prediction*, Organization of Human Brain Mapping (OHBM), Annual Meeting, June 2019.
3. **A. Ghosh***, A. Xifra-Porxas*, G.D. Mitsis, M.H. Boudrias. *Biological brain age prediction using structural MRI: Insights from dimensionality reduction techniques*, International Society for Magnetic Resonance in Medicine (ISMRM), Annual Meeting, May 2019.
4. **A. Ghosh**, F. Dal Maso, G.D. Mitsis, M.H. Boudrias, *Deep Convolutional Architecture with discriminative feature visualization for analysis of EEG data*, Neuroinformatics, August 2018.
5. **A. Ghosh**, Arthur Dehgan, Tarek Lajnef, Raphael Vallat, Jean-Baptiste Eichenlaub, Perrine Ruby, Karim Jerbi, *Unraveling the neural correlates of dream recall: Novel insights from deep convolutional nets*, Neuroinformatics, August 2018.
6. **A Ghosh**, B Bhattacharya, S B R Chowdhury, *AdGAP: Advanced Global Average Pooling*, AAAI-18 Student Abstract and Poster Program (SA-18), February 2018.
7. **A Ghosh**, B Bhattacharya, S B R Chowdhury, *Training Autoencoders in Sparse Domain*, AAAI-18 Student Abstract and Poster Program (SA-18), February 2018.
8. **A. Ghosh**, F. Dal Maso, G.D. Mitsis, M.H. Boudrias, *Deep Semantic Architecture for analysis of motor task EEG data*, Montreal Artificial Intelligence & Neuroscience, November 2017. [**Facebook Best poster award**]
9. F. Dal Maso, B. Desormeau, **A. Ghosh**, M. Roig, M.H. Boudrias, *An acute bout of exercise performed immediately after motor learning alters Beta-band oscillations in the motor cortex*, The Society for Neuroscience, 47th Annual Meeting, November 2017.
10. M. Kassinosopoulos, **A. Ghosh**, E. Dhamala, M.H. Boudrias, G.D. Mitsis. *Cardiac Noise Removal from BOLD fMRI based on a Dynamic Linear Model*, Organization of Human Brain Mapping (OHBM), Annual Meeting, June 2017.
11. **A Ghosh**, B Bhattacharya, S B R Chowdhury, *Handwriting Profiling using Generative Adversarial Networks*, AAAI-17 Student Abstract and Poster Program (SA-17), February 2017. [Link](#)
12. **A Ghosh**, B Bhattacharya, S B R Chowdhury, *SAD-GAN: Synthetic Autonomous Driving using Generative Adversarial Networks*, 30th Annual Conference on Neural Information Processing Systems (NeurIPS) 2016: Deep Learning for Action and Interaction, December 2016.[Link](#)
13. K K Agrawal, **A Ghosh**, *Generative Adversarial Motion to Image Synthesis*, 14th European Conference on Computer Vision (ECCV) 2016 Workshop on Biological and Artificial Vision, Amsterdam, October 2016.

*Equal contribution

TEACHING AND VOLUNTEERING EXPERIENCE

Co-organizer, Neural Information Processing Systems Montreal meetup <i>Montreal Institute for Learning Algorithms (MILA)</i>	Dec 2019
TA, Computational Perception (COMP-546) <i>Dr. Michael Langer, McGill University</i>	Jan - Apr 2019
Reviewer, Reproducibility Challenge <i>International Conference on Learning Representations (ICLR)</i>	Jan 2019
Co-organizer, Deep Learning and Neuroscience reading group <i>Montreal Neurological Institute (MNI)</i>	Dec 2018 - Present
Student Volunteer, HBHL Symposium and Research Day <i>Healthy Brains for Healthy Lives (HBHL), McGill University</i>	9-10 May 2018
Tutor, Deep Learning for Visual Computing Summer School <i>Dr. Debdeep Sheet, IIT Kharagpur</i>	2-8 Jul 2019

SCHOLARSHIPS AND AWARDS

- Integrated Program in Neuroscience (**IPN**) GREAT travel award, June 2019 (\$500)
- Quebec Bio-Imaging Network (**QBIN**) travel award, June 2019 (\$500)
- 2nd prize in McGill **TechIdea** Pitch Competition during the annual John D. Thompson Entrepreneurial Development Seminar: presented “*Telesphorus - brain age prediction system*”, January 2019 (\$250)
- Healthy Brains for Healthy Lives (**HBHL**) Masters fellowship, September 2018 (\$10,000)
- McGill **Faculty of Medicine** Internal fellowship, September 2018 (\$10,000)
- Quebec Bio-Imaging Network (**QBIN**) travel award, June 2018 (\$500)
- Quebec Bio-Imaging Network (**QBIN**) Foreign Students Scholarship, December 2017 (\$7,000)
- Integrated Program in Neuroscience (**IPN**) Recruitment award 2017 (\$5,000)
- MITACS Graduate Fellowship award 2017 (\$15,000) for pursuing graduate studies in Canada
- MITACS Globalink Research Internship award 2016 (\$7,500) for summer internship 2016
- **Honda** Young Engineer and Scientist award, January 2016 (\$3,000)
- **O.P. Jindal** Engineering & Management Scholarship for 2015 & 2016 (\$3,200)
- **Facebook-MAIN** Best Poster award, Montreal AI & Neuroscience Conference, Nov 2017
- **Systems Society Award** for best project work in 8th semester on Systems, May 2017
- Institute **Silver medal** for securing highest CGPA among Electrical engineering graduates, 2017

EXTRA-CIRRICULAR ACTIVITIES

🏆 Best All-rounder Outgoing student (2017)	🎤 Elocution - 2 Gold, 3 Silver (2014-17)
🎭 Governor , Bengali Dramatics Society (2016-17)	🏑 Field Hockey - 1 Bronze (2016)
🎤 Debate - 1 Gold, 1 Silver (2013-15)	💻 IOT-based Hackathon - Gold (2015)
🎭 Dramatics - 10 plays, 2 Gold, 4 Silver (2013-17)	🎤 Director , IIT Kharagpur MUN (2014)
🤖 Robot Maze Traversal event - Best Freshers (2014)	🏆 Best All-rounder fresher (2014-15)